Biochemistry Cumulative Exam

Cytochrome bc_1 is an integral membrane protein in the mitochondrial respiratory chain which transfers two electrons from one molecule of ubiquinol (QH_2) to two molecules of oxidized cytochrome c, and pumps 2 protons from the inside to the outside of the mitochondrial membrane, and delivers two additional protons from QH_2 to the outside of the membrane.

$$QH_2 + 2 \text{ cyt c (oxidized)} + 2H^+_{in} \rightarrow Q + 2 \text{ cyt c (reduced)} + 4\ H^+_{out}$$

Ubiquinol has a redox potential of $E_o' = +0.01$ V, and cytochrome c has a redox potential of $E_o' = +0.26$ V. Under active electron transfer conditions, the electrical potential across the mitochondrial membrane was measured to be $V_{out} - V_{in} = +0.15$ V, and the pH inside and outside the mitochondrial membrane was $pH_{in} = 7.5$, and $pH_{out} = 6.0$.

1. (25) Give a diagram showing how cytochrome bc_1 couples the energy of electron transfer to pumping protons across the mitochondrial membrane, and briefly discuss the mechanism.

2. (25) Calculate the free energy change ΔG^o for electron transfer from 1 mole of ubiquinol to two moles of oxidized cytochrome c. (not including the energy of proton pumping).

3. (40) Give a formula for the free energy change ΔG for pumping protons from the inside to the outside of the mitochondrial membrane. Calculate ΔG for the proton pumping component of the cytochrome bc_1 reaction under the above conditions. You can assume that the energy for transporting two protons from QH_2 to the outside of the membrane is Zero. Therefore, only calculate the ΔG for pumping two protons across the membrane.

4. (10) Calculate how efficiently cytochrome bc_1 couples the exergonic free energy of electron transfer to the endergonic free energy of proton pumping under the above conditions.

Possibly useful formulas:

$\Delta G^o = - RT\ln K_{eq}$; $\Delta G^o = -nF\Delta E_o$; $\Delta G = \Delta G^o + RT\ln([\text{products}]/[\text{reactants}])$.

Membrane Transport: $\Delta G(X_{in} \rightarrow X_{out}) = RT\ln([X_{out}]/[X_{in}]) + ZF(V_{out} - V_{in})$.

Z = charge on X

R = 8.3 Joule/degree.mol, F = 96.485 KJoule/volt.mol, T = 298 °K